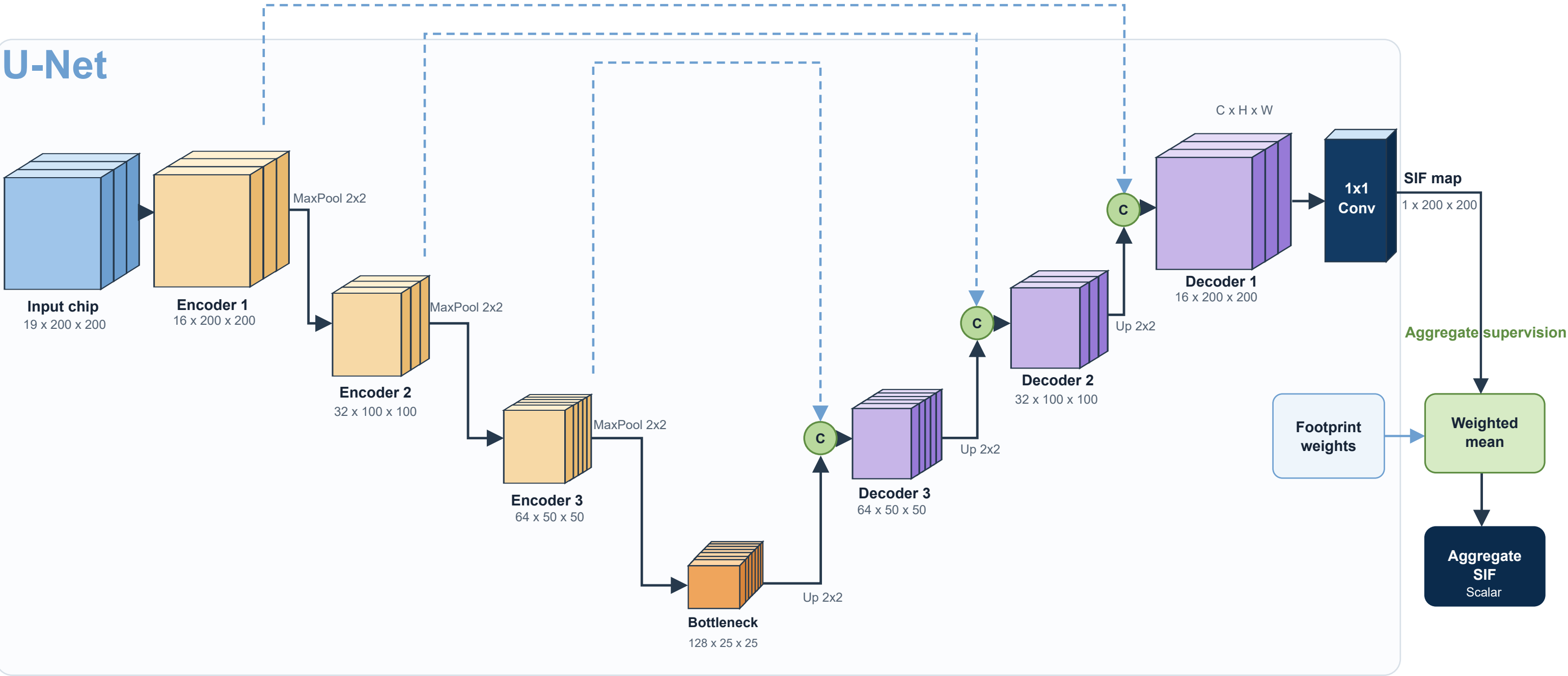


Spatial-Aggregate (4 km) Three-Level U-Net

Three downsamplings | 20 m dense SIF prediction | 485,041 trainable parameters

Skip concatenations

U-Net



Input Composition (19 channels)

8 spectral/biophysical | 9 cover fractions | month sin/cos

Conv block

3x3 Conv → GroupNorm (8) → SiLU x2

Decoder stage

2x2 transposed conv → Concatenate Skip → Conv Block